

Sri Datta Ganesh Bandreddi

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EDUCATION

CARNEGIE MELLON UNIVERSITY

Master of Science in AI Engineering, Biomedical Engineering | GPA: 4.00/4.00

Pittsburgh, PA

Dec 2026

VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY

Bachelor of Technology, Mechanical Engineering | GPA: 8.44/10.00 (Scholar 3.63/4.00)

Nagpur, India

Jul 2022

SKILLS

Languages: Python, C++, Go, Java, SQL, Bash /Shell Scripting

GPU (Acceleration): CUDA, cuDNN, cuBLAS, TensorRT, Triton Inference Server, NCCL, Nsight Systems/Compute

DL Frameworks: PyTorch, JAX, TensorFlow, HF, NeMo/Megatron-LM, LoRA/QLoRA, FP8/BF16 Mixed Precision

Data & ETL: Rapids (cuDF, cuML), NVTabular, DALI, NumPy, SciPy, Pandas, SparkML

Systems & MLOps: Docker, Kubernetes, Slurm, Jenkins/GitHub Actions, MLflow, Git, AWS (ECS, SageMaker), GCP

Data Infrastructure: PostgreSQL, Neo4j, Kafka, Redis, Iceberg, Elasticsearch, FAISS, MongoDB

PROFESSIONAL EXPERIENCE

MOTIONAL

Pittsburgh, PA

Data Platform Intern (Infrastructure)

May 2026 - Aug 2026

- Designed, prototyped, and shipped a metadata-extraction system so teams could learn what is inside an autonomous-vehicle drivelog (about 1 TB each, petabytes in total) before paying to extract the whole thing, delivering a staging PoC and the production design on a medallion architecture
- Made scene-metadata retrieval fast with a new geospatial indexing scheme (H3) and scene embeddings for similarity search, and filed a provisional patent on the approach
- Added an agentic LangGraph layer for natural-language drivelog retrieval behind a FastAPI backend, and benchmarked Ray Data against Spark on Kubernetes to pick the production stack

PHENOM

Hyderabad, India

Product Development Engineer (Enterprise Talent Graph)

Dec 2023 - Jul 2025

- Led six engineers to re-architect a multi-tenant AI analytics platform, refactoring shared services and cutting cross-tenant coupling so it held up under heavy concurrent load
- Grew a production knowledge graph past 1M entities with continuous ingestion, designing validated aggregation pipelines on Neo4j schemas and Iceberg versioned storage
- Owned the Kafka platform (custom exporter, Prometheus scraping, Kafka UI, ZooKeeper-to-KRaft migration) and sped up AI-driven query responses by 40% across 100+ clients by tuning PostgreSQL sharding

BYJU'S

Hyderabad, India

Software Engineer (Core Platform Team)

Jul 2022 - Dec 2023

- Cut infrastructure costs by 70% while keeping 1M+ users well served, deploying scheduled AWS ECS auto-scaling and Terraform IaC tuned to real traffic patterns
- Halved reliance on external identity services by leading development of an in-house auth platform (SAML, SSO, RBAC) in Go and Java (Spring Boot)

PHENOM

Hyderabad, India

Machine Learning Intern

May 2021 - Jun 2022

- Lifted recommendation accuracy by building persona-classification models with spaCy and BERT embeddings across 3 data sources for production recommendation pipelines
- Made recommender preprocessing much faster by moving CPU pipelines to NVTabular on GPU, cutting latency across 2 production databases

PROJECT EXPERIENCE

FLARE: Efficient Attention (Carnegie Mellon University) ([link](#))

Aug 2025 - Present

- Co-authored a linear-attention operator (3rd of 7 authors, submitted to TMLR) later picked up by an NVIDIA and General Motors crash-dynamics effort for roughly 2x lower memory
- Showed over 200x speedups on Long Range Arena (1k to 8k tokens), profiled multi-GPU NCCL training across FP8/BF16 with Nsight Systems, and took FLARE into image classification and diffusion

Distributed LLM Fine-Tuning with Megatron-LM (Carnegie Mellon University)

Aug 2025 - Oct 2025

- Fine-tuned a 7B-parameter LLM to stable convergence with about 40% less memory on four H100 GPUs using tensor and pipeline parallelism, gradient checkpointing, and LoRA/QLoRA adapters
- Pushed training throughput to 2.1x over naive data parallelism by tuning NCCL, gradient bucketing, and optimizer-state sharding

CT-Transformer, Vision Transformers (Carnegie Mellon University)

Sep 2025 - Dec 2025

- Hit 96.22% macro AUC-ROC on 15-class VinBigData chest X-ray classification, beating ResNet-50, EfficientNet-B3, and ViT baselines with a 210M-parameter Swin Transformer
- Sped up inference 3x over PyTorch eager by converting to TensorRT (FP16/INT8), speeding preprocessing with DALI, and serving on Triton with adaptive batching